# Bhaskar Roberts

Postdoctoral Researcher in Computer Science

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## Education

**Princeton University** — *Princeton, NJ*

**BSE in Electrical Engineering** — *June 2019*

- Minor in Applied & Computational Math
- Summa Cum Laude
- Activities: CS tutor, peer academic advisor, SEAS tour guide, opinion writer for the Daily Prince, musician for Triangle Club

**University of California, Berkeley** — *Berkeley, CA*

**PhD in Computer Science** — *August 2026*

- Research on cryptography and quantum information
- Supported by an NDSEG Fellowship
- Activities: council president of the UC Choral Ensembles, business manager of Noteworthy A Cappella

## Research

***Certified Randomness with Optimal Rate*** — *Preprint*

Siddhartha Jain, Saachi Mutreja, Bhaskar Roberts

***Unconditional Certified Randomness without Structure*** — *Preprint*

Andrea Coladangelo, Dakshita Khurana, Saachi Mutreja, Bhaskar Roberts, Joseph Slote, Avishay Tal

***Query-Limited RAM Programs and their Applications*** — *Preprint*

Jiahui Liu, Justin Raizes, Bhaskar Roberts, Omri Shmueli

***On Best-Possible One-Time Programs*** — *Preprint*

Aparna Gupte, Jiahui Liu, Luowen Qian, Justin Raizes, Bhaskar Roberts, Mark Zhandry

***Threshold Anonymous Credentials with Silent Setup*** — *Preprint*

By Preshtha Garg, Sanjam Garg, Guru-Vamsi Policharla, Bhaskar Roberts

***Collaborative zkSNARKs with Sublinear Prover Time and Constant Proof Size*** — *Preprint*

By Zhiyong Fang, Sanjam Garg, Bhaskar Roberts, Wenxuan Wu, and Yupeng Zhang

***Certified Randomness without Structure Against Shallow-Query Adversaries*** — *FOCS 2026*

Dakshita Khurana, Bhaskar Roberts, Avishay Tal

***Collaborative Incrementally Verifiable Computation*** — *TCC 2026*

Eden Aldema Tshuva, Sanjam Garg, Abhiram Kothapalli, Rotem Oshman, Omkant Pandey, Bhaskar Roberts

***ANARKey: A New Approach to (Socially) Recover Keys*** — *EuroS&P 2026*

by Aniket Kate, Pratyay Mukherjee, Hamza Saleem, Pratik Sarkar, and Bhaskar Roberts

***Malicious Security in Collaborative zk-SNARKs: More than Meets the Eye*** — *Crypto 2025*

by Sanjam Garg, Aarushi Goel, Abhishek Jain, Bhaskar Roberts, and Sruthi Sekar

***Unconditional Quantum Cryptography with a Bounded Number of Keys*** — *CiC Volume 2, Issue 1*

by Vipul Goyal, Giulio Malavolta, and Bhaskar Roberts

***Quantum One-Time Programs, Revisited*** *STOC 2025*

by Aparna Gupte, Jiahui Liu, Justin Raizes, Bhaskar Roberts, and Vinod Vaikuntanathan

***Software with Certified Deletion*** *QIP 2023, Eurocrypt 2024*

by James Bartusek, Vipul Goyal, Dakshita Khurana, Giulio Malavolta, Justin Raizes, and Bhaskar Roberts

***Franchised Quantum Money*** *Asiacrypt 2021*

by Bhaskar Roberts and Mark Zhandry

***Security Analysis of Quantum Lightning*** *Eurocrypt 2021*

by Bhaskar Roberts

- Adapted from my undergraduate thesis, which earned an *A+* and the *Peter Mark Prize*.

***Efficient Algorithms for QSPD*** *ERN Conference in STEM 2020*

by Andrew Childs, Bhaskar Roberts, Brennan Schaffner, and Finn Voichick

- Won *Best Poster in Computer Science*.

## Work Experience

**NTT Research** *Sunnyvale, CA*

Postdoctoral Researcher *August 2026 - Present*

- Research on cryptography and quantum information

**NTT Research** *Sunnyvale, CA*

Research Intern *Spring 2025*

- Research on quantum certified randomness. Work published at FOCS 2026.
- Worked with Dakshita Khurana.

**Supra Research**

Research Intern *Summer 2024*

- Designed a new algorithm for account key recovery and proposed a new model of security.
- Worked with Pratyay Mukherjee, Pratik Sarkar, and Aniket Kate.

**NTT Research** *Sunnyvale, CA*

Research Intern *Summer 2023*

- Developed constructions and techniques for unconditional quantum cryptography.
- Worked with Vipul Goyal.

**Nvidia** *Santa Clara, CA*

ASIC Design Intern *Summer 2018*

- Built machine learning models to predict and optimize power usage on Nvidia GPUs.

**NASA Langley Research Center** *Hampton, VA*

Hardware Engineering Intern *Summer 2017*

- Designed, built, and tested an optical transceiver for a CubeSat network.
- Designed amplifier circuits and a printed circuit board, and programmed an FPGA.

## Honors

**Outstanding GSI Award** Teaching award given to 10% of graduate student instructors (GSIs), UC Berkeley, 2025

**Best Poster in CS** ERN Conference in STEM 2020

**Peter Mark Prize** Awarded to two seniors for outstanding senior thesis research in electrical engineering, Princeton University, 2019

**Phi Beta Kappa** Awarded to the top 10% of Princeton's graduating class, Princeton University, 2019

**Tau Beta Pi** Served as chapter president of the engineering honor society, Princeton University, 2017

**Shapiro Prize** Awarded to 3% of juniors to recognize outstanding academic achievement in the past year, Princeton University, 2017